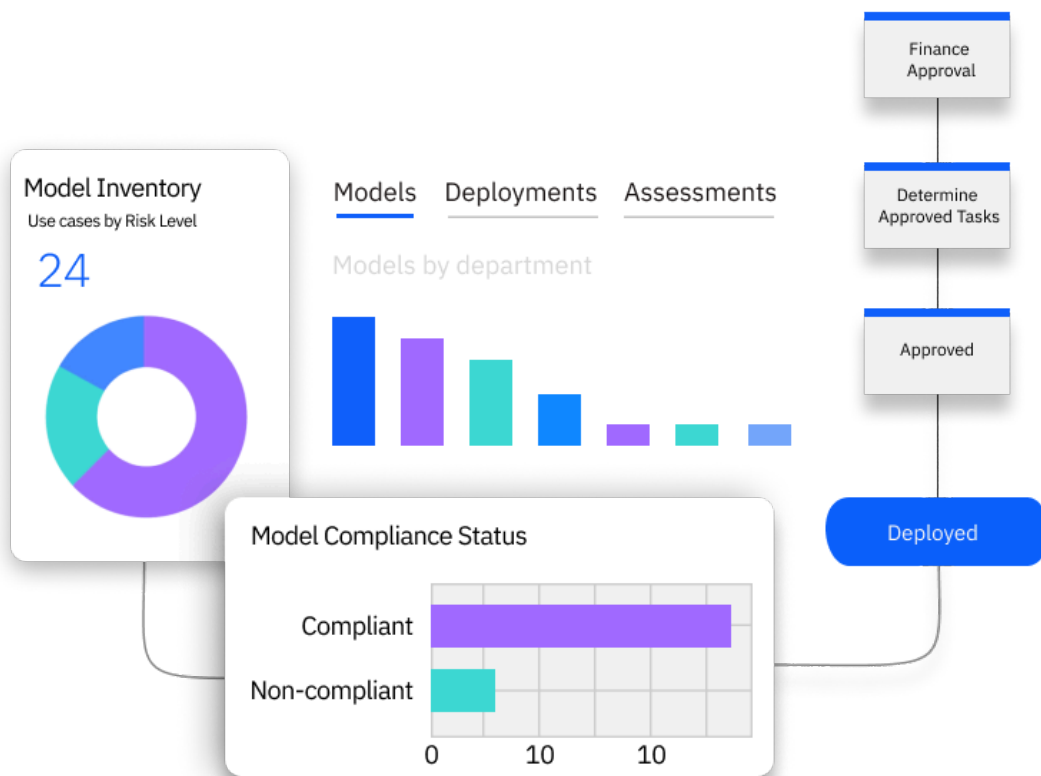


Tracing and monitoring of agentic applications in watsonx

Hands-on Lab Guide



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Tracing and monitoring of agentic applications in watsonx

Introduction

Agentic AI refers to artificial intelligence systems that can autonomously plan, make decisions, and take actions to achieve specific goals — with minimal or no human intervention. These systems are not just reactive; they are goal-oriented, capable of reasoning learning, and adapting over time. This makes them ideal for organizations looking to automate processes, reduce costs, and add new capabilities for their customers. However, agentic AI adds significant complexity for those trying to govern the models and prompts to ensure that the output is trustworthy, fair, and accurate.

IBM watsonx.governance is a core component of watsonx, IBM's enterprise-ready AI and data platform that's designed to multiply the impact of AI across an enterprise. It offers tools for tracing, monitoring, and evaluating agentic AI.

About this hands-on lab

The main objective of this lab is to experiment with the watsonx.governance tools for monitoring agentic AI. Users will interact with prompts via Jupyter notebook, and see the kinds of evaluations and metrics available.

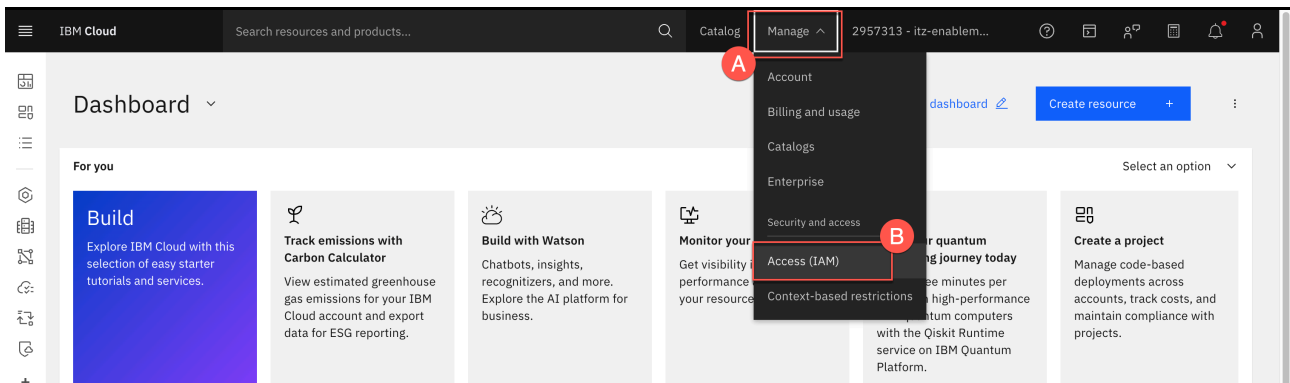
Signing into the environment and gathering credentials

Your lab environment contains all the necessary services for running and monitoring agentic AI, including watsonx.ai and watsonx.governance. In this section, you will sign in to the lab and gather the necessary credentials to access these services.

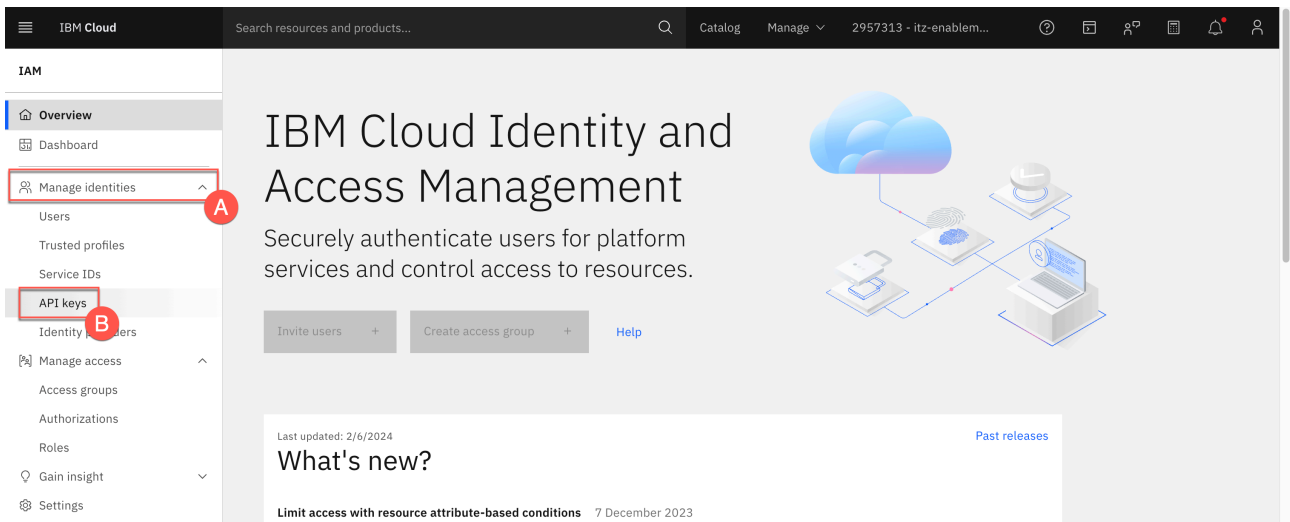
Create an API key

Your first task will be to create an API key that will allow you to authenticate with the watsonx.ai and watsonx.governance services.

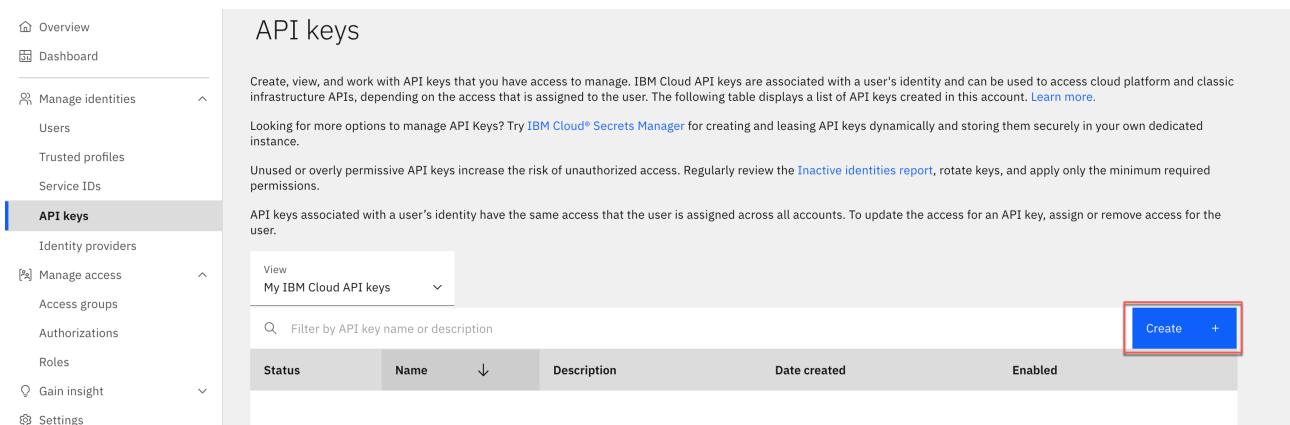
1. Sign in to the IBM Cloud environment using the link and credentials provided in the lab materials.
2. Click on the [Manage](#) menu (A) to open it, then click on the [Access \(IAM\)](#) menu item (B). The [IBM Cloud Identity and Access Management](#) screen opens.



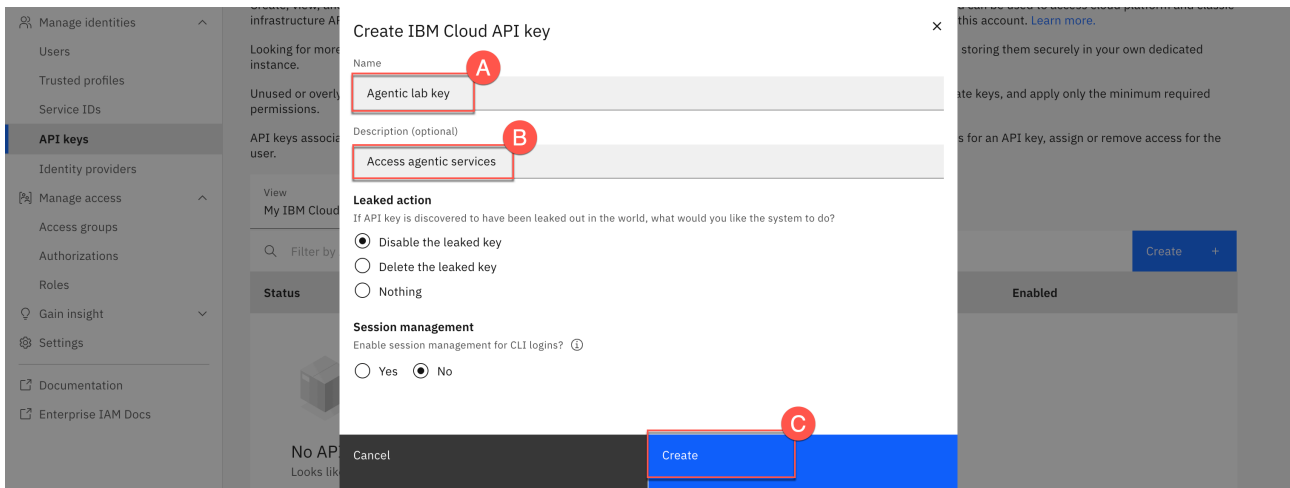
- From the menu on the left, click on the [Manage identities](#) menu item (A) to expand it if necessary. Then click on the [API keys](#) menu item (B).



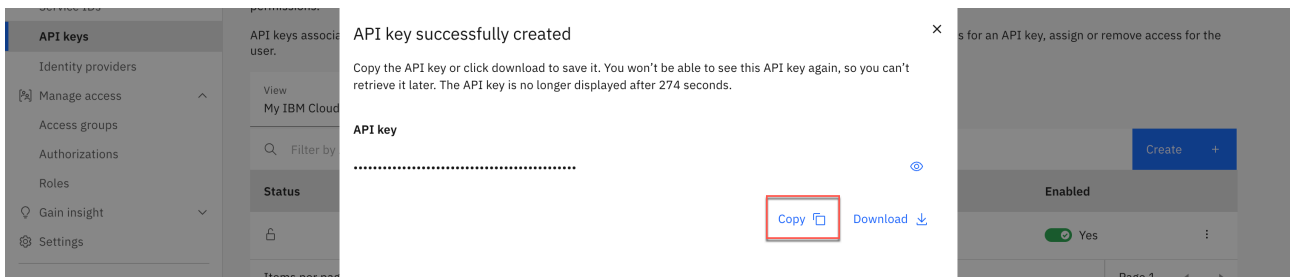
- Click on the [Create](#) button to create a new API key. The [Create IBM Cloud API key](#) modal window opens.



- Give your key a [name](#) (A) and an optional [description](#) (B). Click on the [Create](#) button (C) to create the key.



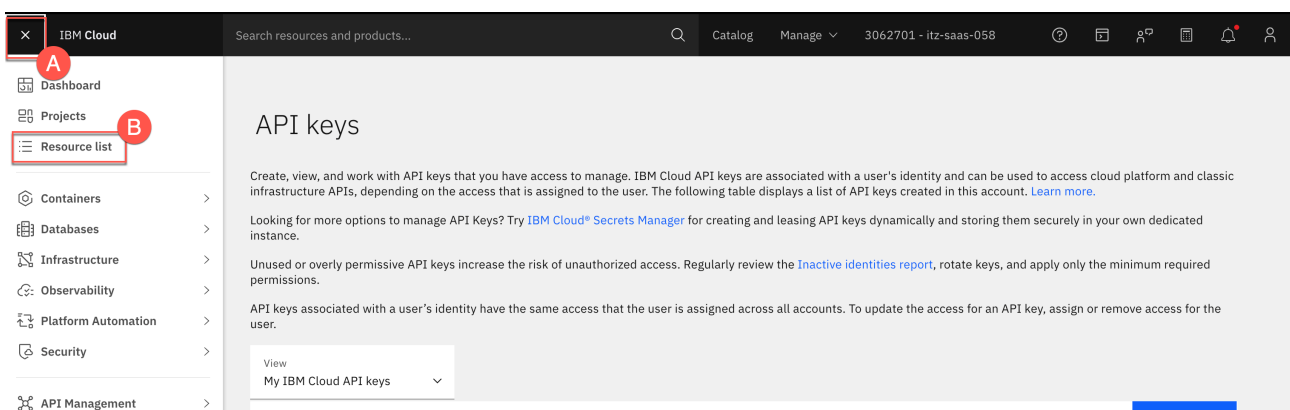
- Click on the [Copy](#) button (A) to copy the key to your clipboard, then click on the close button to close the modal window. Note that you will be unable to access the key after you close the modal window, so be sure to copy it to your machine's clipboard before closing it. If you do not copy your key, you can delete it and create a new one. Paste the key into a text editor on your local machine.



You now have the credentials necessary to proceed with the lab.

Launch the governance service

- Click on the [Navigation menu](#) button (A) to expand it, then click on the [Resource list](#) button (B). The resource table opens.



- From the resource list table, click on the [AI / Machine Learning](#) section (A) to expand it. Locate the [watsonx.governance](#) service from the list (B) and click on the linked name to open it.

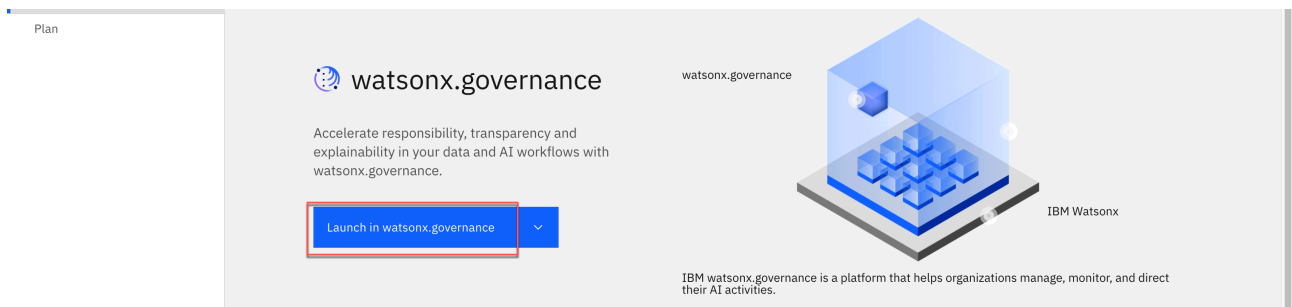
 If you have more than one [watsonx.governance](#) service, choose the service in [Dallas \(us-south\)](#).

Resource list

Create resource +

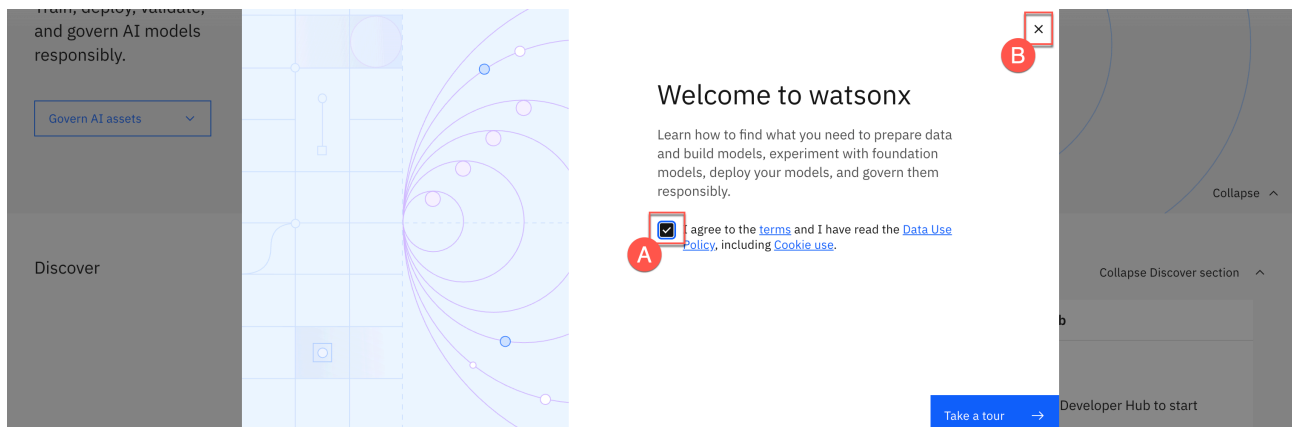
Name	Group	Location	Product	Status	Tags
Filter by name or IP address... Filter by group... Filter... Filter... Filter... Filter...					
AI / Machine Learning (3)					
gov-550000w6t0	eid-68ca2705f262a7ad7bad874b	Dallas (us-south)	watsonx.governance	Active	eid:68c... +1
wml-550000w6t0	eid-68ca2705f262a7ad7bad874b	Dallas (us-south)	watsonx.ai Runtime	Active	eid:68c... +1
ws-550000w6t0	eid-68ca2705f262a7ad7bad874b	Dallas (us-south)	watsonx.ai Studio	Active	eid:68c... +1
Analytics (1+)					
Application development (0)					
Asset management (0)					

3. Click on the [Launch in watsonx.governance](#) button to launch the service in a new browser tab.



4. If the [Provide your information to continue](#) screen opens, click the [Continue](#) button.

5. Check the box in the [Welcome to watsonx](#) popup (A) to agree to the terms and conditions. Then click the [close](#) button (B) to close the popup. If the [Dive deeper](#) popup window appears, close it as well.

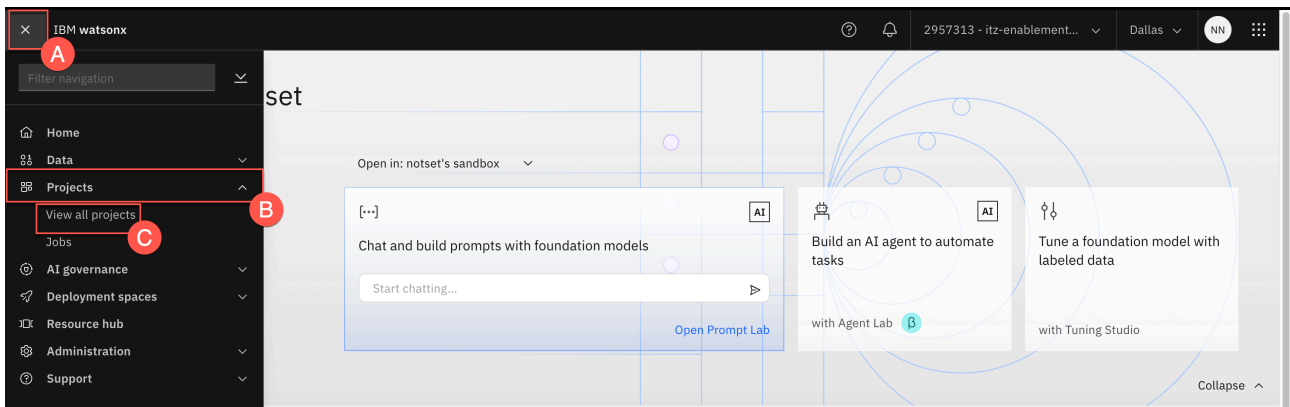


Work with agentic AI code

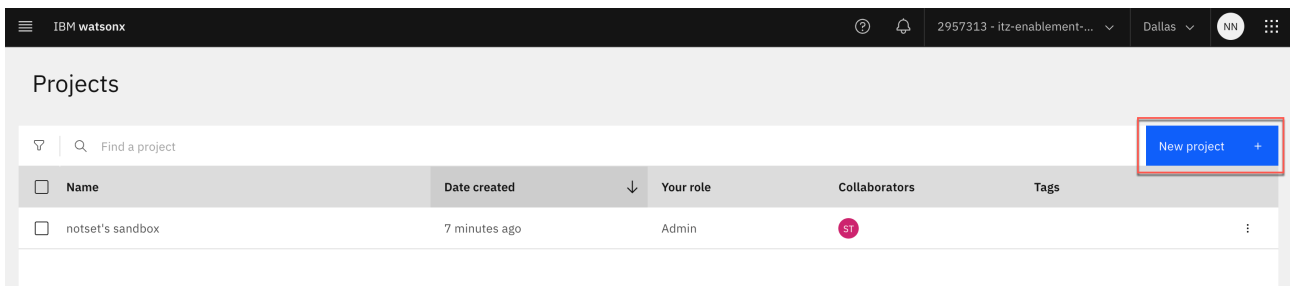
In this section of the lab, you will work with a Jupyter notebook in a watsonx project. Projects are collaborative environments with full access management that allow data scientists, data engineers, AI engineers, and subject-matter experts to work together on AI and machine learning initiatives. They can contain code, data, connections to data sources, AutoAI experiments, and more.

Create a watsonx project

1. Click on the [Navigation menu](#) (A) to open the menu. Then click on the [Projects](#) menu item (B) to expand it. Finally, click on the [View all projects](#) menu item (C). The projects screen opens.



2. Click on the [New project](#) button. The [Create a project](#) screen opens.



3. Give your project a [name](#) and click on the [Create](#) button to create the project.

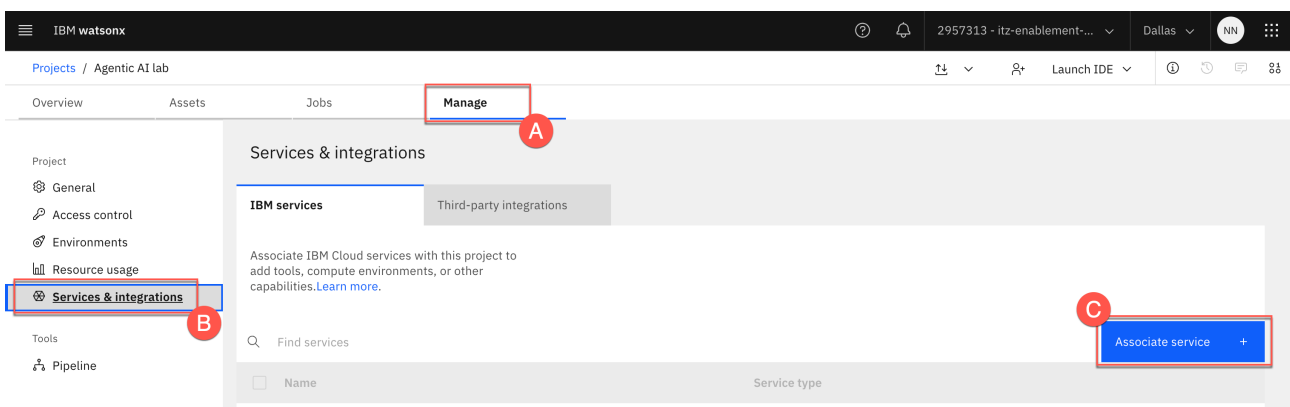
Create a project

Start with a new, blank project or select from where to import an existing project.

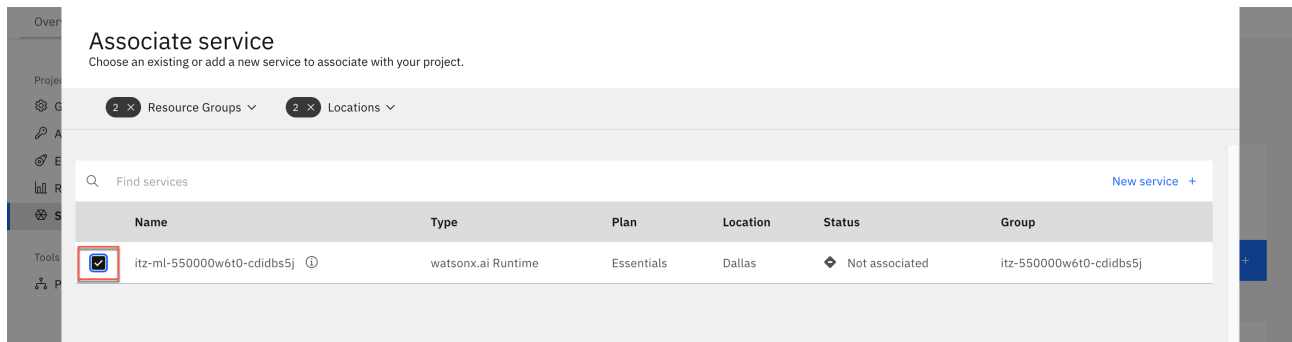


After your project has been created, you will need to associate your project with the watsonx.ai service in order to be able to leverage its features.

4. Click on the [Manage](#) tab (A). Click on the [Services & integrations](#) menu item (B). Then, click on the [Associate service](#) button (C). The [Associate service](#) window opens.



5. Check the box to the left of the [watsonx.ai Runtime](#) service.



- Click on the [Associate](#) button to associate the watsonx.ai service with your project and close the window.

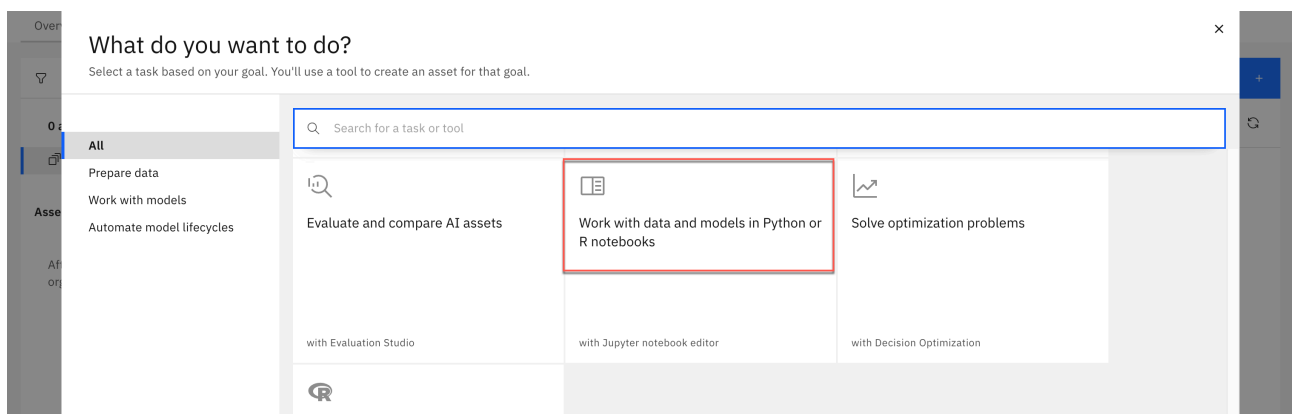
Import the lab notebook

In this section, you will work with the lab notebook to interact with and govern your agents.

- From your project, click on the [Assets](#) tab (A). Click on the [New asset](#) button (B). The [What do you want to do?](#) modal window opens.

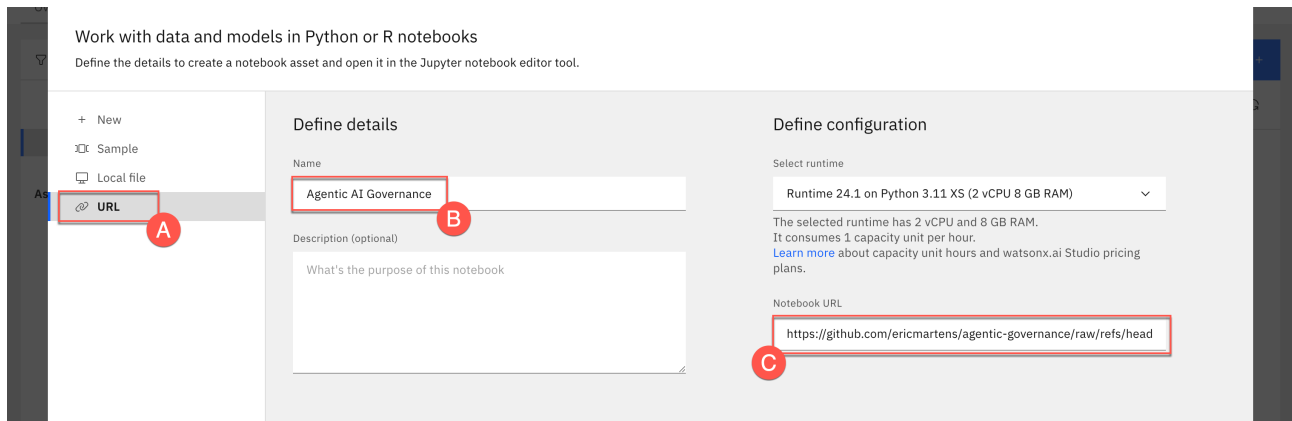


- Scroll down to the [Work with models](#) section and click on the [Work with data and models in Python...](#) tile.



- Click on the [URL](#) tab (A). Give your notebook a [name](#) (B), then copy and paste the following URL into the [Notebook URL](#) field (C):

<https://github.com/ericmartens/agentive-governance/raw/refs/heads/main/governed-agentive-lab.ipynb>

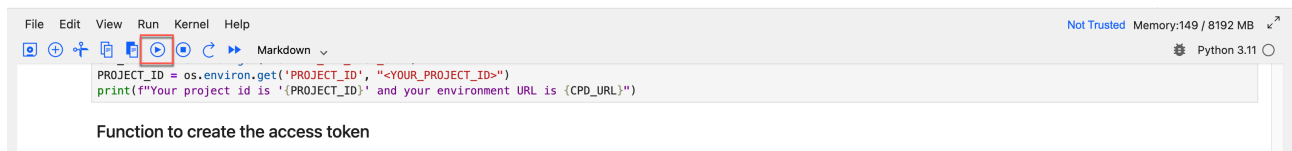


4. Click on the **Create** button. The notebook editor will open and load the notebook from the URL.

The notebook consists of two types of cells: markdown cells, which contain explanatory text, and code cells like the one below:



Notebooks are designed to be run one cell at a time from top to bottom. You can click on a cell to highlight it and make it active. You can run an active cell by pressing the **Shift + Enter** keyboard shortcut, or by clicking on the run button in the task bar:



Running the active cell will execute any code in the cell, and will make the following cell active. If the cell contains only markdown text, no action will be taken and the next cell will become active. As a cell runs, an asterisk (*) will appear in the brackets to the left of the cell. When it completes, the asterisk will be replaced by a number. You can run active cells while code is still executing; they will execute in the order you ran them when the previous cell is finished, unless interrupted by an error.

5. Upon being executed, one of the cells will request the watsonx API key that you created earlier in the lab. Paste that value into the field, then continue executing the cells.
6. Run through the cells in the notebook one at a time, taking a moment to read and understand the text above each cell. The notebook will install Python libraries used to work with agentic AI. It will then set up IBM's Granite language model for text generation, binding it to two very basic tools. Finally, it establishes a second LLM (llama) as a judge in order to compute relevance and accuracy metrics.

Follow the instructions in the notebook to complete the lab.